

Forecast-Driven Energy-Aware Scheduling for Smart Manufacturing

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Abstract—Energy-aware production scheduling is becoming increasingly important as manufacturers face time-varying electricity prices, peak-demand charges, and carbon-reporting requirements. This paper presents a forecast-driven scheduling framework that combines probabilistic energy forecasting with production scheduling and constraint-programming refinement. The system uses tariff-weighted quantile XGBoost models to estimate energy use, carbon emissions, and power factor, followed by six scheduling strategies: FCFS, EDF, a proposed Forecast-Driven Priority Dispatching with Tariff Shifting (FD-PDTS) heuristic, a robust FD-PDTS variant, a makespan-oriented baseline, and a warm-started CP-SAT refinement method. A twelve-metric KPI layer evaluates economic, operational, compliance, environmental, and quality outcomes. On a single synthetic benchmark containing 180 jobs and eight machines, the hybrid schedule reported a reduction in energy cost and peak grid load relative to FCFS, at the cost of increased average waiting time. However, a post-hoc implementation audit identified unresolved defects in recursive forecasting, quantile calibration, power-factor bounds, and peak-load unit consistency that directly affect these reported figures, and the benchmark itself relies on a single synthetic instance. The paper therefore presents the system as a prototype methodology rather than validated industrial evidence, and documents the specific corrections required before the numerical results can support stronger performance claims.

Index Terms—smart manufacturing, energy-aware scheduling, quantile forecasting, predict-then-optimize, XGBoost, CP-SAT, demand response

I. INTRODUCTION

Manufacturing schedules are normally built around production constraints such as machine capacity, processing time, job priority, and deadlines. In facilities exposed to time-of-use electricity prices or demand charges, however, the time at which a flexible job is processed can also affect operating cost and grid demand. This creates a scheduling problem in which production decisions and energy conditions should be considered together.

Energy-aware scheduling has been studied using mathematical programming, dispatching rules, heuristics, and hybrid optimization methods [1], [2]. At the same time, modern manufacturing systems increasingly use data-driven models to estimate future energy conditions [3]. The challenge is that the scheduling model may depend on quantities that are not known in advance. A useful way to connect these two stages is the predict-then-optimize paradigm, where a predictive model supplies estimates that are subsequently consumed by an optimization model [4].

This work implements that idea as an end-to-end software pipeline. Historical industrial energy records are processed to generate temporal features, quantile XGBoost models produce forecasts, and a scheduling engine uses those forecasts together with job and machine information. A heuristic schedule is then supplied as a warm start to CP-SAT for further refinement. The implementation also includes KPI computation and an interactive dashboard.

The main contributions are:

- an end-to-end forecast-driven architecture for energy-aware production scheduling;
- the FD-PDTS tariff-aware dispatching heuristic and its uncertainty-aware variant;
- integration of a heuristic solution with a warm-started CP-SAT refinement stage;
- a unified KPI framework covering cost, production, compliance, emissions, and power quality; and
- an explicit validity assessment that separates preliminary prototype results from claims that require further experimental validation.

II. RELATED WORK

Energy-efficient scheduling has been reviewed extensively, including formulations that incorporate electricity consumption, peak demand, and carbon objectives [1], [2]. Time-of-use pricing has also been incorporated into flexible job-shop scheduling, showing that production operations can sometimes be shifted toward lower-cost periods without violating operational constraints [5]. More recent formulations combine several energy objectives with constraint programming [6].

The forecasting stage of this work is based on quantile regression. Quantile methods provide information about different points of a predictive distribution rather than only a single expected value [7]. XGBoost is used because gradient-boosted trees are effective for structured tabular data and provide an efficient implementation for nonlinear feature interactions [8]. The broader predict-then-optimize formulation provides the conceptual connection between forecasting and downstream decisions [4].

Constraint programming is used as the final refinement mechanism. CP and mixed-integer formulations have been applied to flexible job-shop scheduling with energy objectives [9], [10]. Reinforcement-learning approaches provide another direction for adaptive scheduling, but they generally require

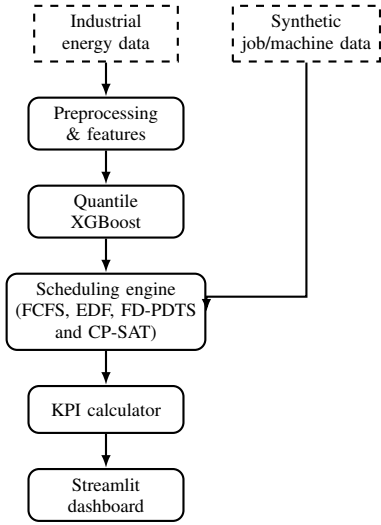


Fig. 1. End-to-end architecture of the forecast-driven scheduling framework.

more training data and can be less transparent than explicit optimization pipelines [11], [12].

III. SYSTEM ARCHITECTURE

The framework has four stages: data preparation, forecasting, scheduling, and evaluation. The energy dataset supplies historical observations for forecasting, while synthetic job and machine tables describe the production problem. Forecast outputs are passed to the scheduling engine, and every candidate schedule is evaluated by the same KPI layer.

The implementation is organized into modules for preprocessing, forecasting, scheduling, KPI calculation, synthetic-data generation, and the Streamlit dashboard. Configuration values such as the random seed, tariff levels, scheduling resolution, and forecast settings are centralized so that experiments use a consistent configuration.

A. Data

The energy data contains 35,040 observations sampled every 15 minutes over 2018. It includes active energy usage, reactive-power measurements, carbon information, power factor, weekday/weekend status, day-of-week, and load type. The scheduling benchmark contains 180 jobs assigned across eight machines. Jobs have processing durations, deadlines, priorities, setup types, and machine compatibility constraints.

B. Preprocessing and Features

Energy records are timestamp-normalized, sorted chronologically, checked for duplicates, and converted to appropriate numerical and categorical types. Missing values are imputed using forward/backward filling. Calendar features are extracted and cyclically encoded:

$$x_{\sin} = \sin(2\pi x/P), \quad x_{\cos} = \cos(2\pi x/P), \quad (1)$$

where P is the relevant period.

Lag features at 1, 2, 4, and 96 samples represent recent and daily history. Rolling statistics over 4 and 96 samples

TABLE I
SUMMARY OF EXPERIMENTAL DATA.

Item	Value
Energy records	35,040
Sampling interval	15 min
Energy coverage	1 year
Jobs	180
Machines	8
Average job duration	62.75 min
Peak-hour arrivals	75%
Compatible machines/job	2–4
Setup types	3

capture short- and long-term variation. The data is divided chronologically into 60% training, 20% validation, and 20% test sets to avoid temporal leakage.

IV. FORECASTING MODEL

The forecasting component produces p10, p50, and p90 estimates for active energy and carbon emissions, together with a p50 estimate for power factor. Seven XGBoost models are therefore trained independently.

A. Tariff-Weighted Quantile Objective

Quantile regression minimizes the pinball loss. For quantile τ and sample weight w_i ,

$$L_\tau = \sum_i w_i \begin{cases} \tau(y_i - \hat{y}_i), & y_i \geq \hat{y}_i, \\ (1 - \tau)(\hat{y}_i - y_i), & y_i < \hat{y}_i. \end{cases} \quad (2)$$

The implementation gives higher weights to maximum-load periods because forecasting errors during expensive periods have greater downstream scheduling consequences. The configured weights are 3.0 for maximum load, 1.5 for medium load, and 1.0 for light load.

The common XGBoost configuration uses 300 estimators, maximum depth 5, learning rate 0.05, and 0.8 row and column subsampling, with random seed 42. Models are stored using Joblib for reuse by the scheduling and dashboard layers.

The power-factor model in particular is later found to violate its physical bounds; see Section VII for details and required corrections before these forecasts are used in reporting.

V. ENERGY-AWARE SCHEDULING

Let J be the set of jobs and M the set of machines. Each job must be assigned to a compatible machine and a feasible start time. Jobs on the same machine cannot overlap, and setup requirements are included when the sequence changes. The scheduling objective balances energy cost, waiting time, and tardiness:

$$\min \alpha C + \beta W + \gamma T, \quad (3)$$

where C , W , and T denote energy cost, waiting time, and tardiness.

A. Scheduling Strategies

Six strategies are evaluated.

FCFS orders jobs by arrival time and schedules them at the earliest available machine slot. It is a simple tariff-unaware baseline.

EDF orders jobs by deadline and attempts to complete them before their due dates. It provides a stronger deadline-oriented baseline but does not consider electricity prices.

FD-PDTS is the proposed tariff-aware heuristic. Jobs are prioritized using deadline slack and job priority. For each compatible machine, feasible start times are evaluated using the p50 forecast. The selected placement minimizes a score combining forecast-based energy cost, waiting time, tardiness, and a peak-load guard.

FD-PDTS Robust replaces the p50 tariff estimate with the upper forecast band when scoring candidate placements. This is intended to make the heuristic less sensitive to underestimated energy requirements, although the current implementation is only a heuristic robustness bias rather than a formal robust-optimization model.

Makespan Greedy orders jobs by processing duration and chooses the machine with the earliest completion time. It is intended as a speed-oriented baseline.

Hybrid CP-SAT first obtains a feasible FD-PDTS schedule and then supplies it as a warm start to an OR-Tools CP-SAT model. The solver refines start times and machine assignments under precedence, capacity, deadline, and setup constraints.

B. Machine Availability

Machine availability is represented by an interval-based calendar rather than a single “next free” pointer. This allows the scheduler to use gaps between shifts and other occupied intervals. For a candidate placement, the calendar searches for the earliest contiguous gap that can accommodate processing and the required setup.

VI. EVALUATION

The system evaluates schedules using economic, operational, compliance, environmental, and quality indicators. Important measures include total energy cost, peak grid load, makespan, utilization, average waiting time, idle time, on-time completion, tardiness, carbon emissions, and power-factor penalty.

On the reported benchmark, the hybrid schedule reduces energy cost from INR 186,096 to INR 125,833, a reduction of 32.4%. Reported peak load falls from 778.28 kW to 448.28 kW, or 42.4%. Carbon emissions also fall from 579.15 to 411.78 tCO₂. These gains come with a tradeoff: average waiting time increases from 54.6 to 94.5 minutes, while makespan increases slightly from 49.25 to 49.50 hours.

The forecasting test results are less satisfactory. The p50 energy model reports an MAE of 19.85 kWh and RMSE of 35.65 kWh. Its MAPE is 32.24%. Quantile coverage is substantially different from the nominal levels: p10 coverage is 99.99% and p90 coverage is 48.46%. Quantile ordering is also violated in 13.14% of test rows.

TABLE II
BENCHMARK RESULTS ON THE SINGLE SYNTHETIC INSTANCE.

Metric	FCFS	EDF	FD-PDTS	Hybrid
Cost (INR)	186096	185686	167723	125833
Peak load (kW)	778.28	778.28	778.28	448.28
Makespan (h)	49.25	49.00	49.25	49.50
Utilization (%)	47.78	48.02	47.78	47.54
Waiting (min)	54.6	61.2	89.3	94.5
Delay (min)	0	0	0	0
On-time (%)	100	100	100	100
CO ₂ (t)	579.15	606.38	521.39	411.78

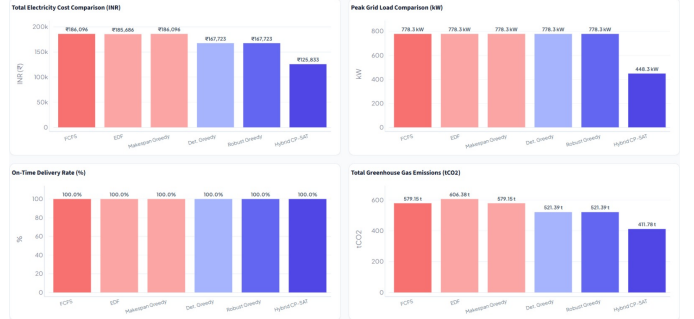


Fig. 2. Dashboard comparison of scheduling strategies.

VII. VALIDITY ASSESSMENT

The reported scheduling results should not be interpreted as validated industrial evidence. The implementation audit identified four important issues.

First, the 48-hour recursive forecast updates only the one-step lag. Longer lags, rolling statistics, and the exponentially weighted moving average remain fixed instead of being recomputed from previous predictions. Consequently, the long-horizon forecast is not currently reliable.

Second, independently trained quantile models produce crossing intervals in 13.14% of test rows. Quantile ordering should satisfy $\hat{y}_{0.10} \leq \hat{y}_{0.50} \leq \hat{y}_{0.90}$. Monotone rearrangement is one possible correction [13].

Third, the power-factor model produces a constant value of 8.0 even though power factor is physically bounded near $[0, 1]$. The target representation or model output therefore requires correction.

Fourth, the peak-load KPI mixes kWh forecast quantities with kW power values without the required 15-minute conversion. A 15-minute energy value corresponds to power through $P = E/0.25$ hours. The KPI must therefore be recomputed with consistent units before publication.

The benchmark itself is also deliberately favorable to energy-aware scheduling: 75% of arrivals occur during peak tariff periods, machines have heterogeneous power characteristics, and deadlines leave some room for shifting. In addition, the makespan baseline is currently numerically identical to FCFS, while the deterministic and robust FD-PDTS variants

produce identical benchmark results. These observations limit the strength of the current comparison.

VIII. DISCUSSION

The implementation demonstrates that forecasting and scheduling can be connected through a practical software pipeline rather than treated as independent modules. The most useful engineering decision was the interval-based machine calendar, which represents production availability more realistically than a single machine pointer. Warm-starting CP-SAT also provides a practical way to combine a fast heuristic with an exact constraint solver under a fixed time budget.

At the same time, the current results show why an end-to-end system must be validated at every stage. A forecasting defect can propagate into the scheduler, and a unit error in a KPI can make an apparently precise result misleading. For this reason, future experiments should include automated checks for quantile ordering, recursive forecasting, physical bounds, schedule feasibility, and KPI dimensional consistency.

IX. FUTURE WORK

The immediate priority is to correct the recursive forecast and validate the custom quantile objective against a standard quantile-loss implementation. Forecasts should also be compared with seasonal-naive, linear-regression, and standard XGBoost baselines. Power-factor predictions should be constrained to their physical range.

On the scheduling side, the robust FD-PDTS formulation should be replaced or complemented by an explicit robust or distributionally robust objective. The CP-SAT objective should directly include carbon, peak-load, and power-factor terms. Setup and machine-availability assumptions should be identical across all baselines.

Most importantly, the benchmark should be expanded to many independently generated instances with different job counts, machine fleets, deadline tightness, peak-load proportions, and tariff spreads. Results should include confidence intervals, solver runtime, optimality gaps, and fixed random seeds. Additional baselines such as SPT, EDD, CP-SAT without warm start, mixed-integer programming, and an oracle schedule with perfect future information would provide a stronger evaluation.

CODE AND DATA AVAILABILITY

The implementation (preprocessing, forecasting, scheduling, KPI calculation, and dashboard modules) is publicly available at <https://github.com/hemanth1139/Smart-Machine-Scheduling-for-Energy-and-Process-Optimization>. The repository includes the synthetic benchmark generator; corrections described in Section VII are being tracked and applied there.

X. CONCLUSION

This paper presented a prototype predict-then-optimize framework for energy-aware manufacturing scheduling. The system combines tariff-weighted quantile forecasting, tariff-aware dispatching, warm-started CP-SAT refinement, and a

common KPI evaluation layer. On the current synthetic benchmark, the hybrid method reports substantial reductions in energy cost and peak load compared with FCFS, but these improvements are accompanied by increased waiting time.

The implementation audit also identified forecasting, physical-bound, and KPI-unit problems that must be corrected before the results can support industrial claims. The main value of the present work is therefore the integrated methodology and reproducible engineering structure rather than the numerical performance of a single benchmark. Multi-instance experiments and corrected forecasting and evaluation components are necessary next steps.

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